Common incumbents and row-scaling attribution

Focused exploration for OMS — 15 October 2026

Question. How much of the observed external-scaling benefit survives when Base and GM receive the same feasible incumbent? This addresses mechanism attribution in the existing paper, without introducing cuts, decomposition, prediction or a new preconditioner.

1. Controlled intervention

Three known SG–Ter–Mer cases retain 6,120 variables, 10,080 rows and 1,440 binaries each. Cases 001 and 003 previously favored GM; case 042 favored Base. This is retrospective diagnosis in one family, not held-out validation. Base/GM × Cold/CommonStart × seeds 503,509,521 gives 36 sequential MILP calls. Gurobi 13.0.2 uses four threads, 12 seconds/12 work units, gap 10^{-3}, primal tolerance 10^{-6}, integer tolerance 10^{-5}, normal presolve/internal scaling and IntegralityFocus=0. Frozen exact row factors are reused; objectives, domains and the underlying mathematical models are fixed. This auxiliary MPS/power-of-two bank differs from the historical LP campaign; it does not establish the cause of that campaign's reported work reduction.

For each case, only the integer pattern of archived GM seed 307 is used. SoPlex produces an exact feasible continuation and a matching dual bound for the fixed-pattern LP. This certifies that LP, not global MILP optimality. The rational point is converted to one binary64 vector supplied identically to both policies. All 18 starts are loaded and their pattern/cost is observed. Original/exported residuals from conversion are retained separately.

2. Completion and work

Each cell gives native completions out of three and median native work over all three calls, including censored ones. Ratios of censored runs are not interpreted as speedups; native completion is not a certified MILP gap.

Case	Initialization	Base: completes; work	GM: completes; work
001	Cold	0/3; 12.0001	3/3; 5.4909
001	CommonStart	3/3; 6.4158	3/3; 6.7719
003	Cold	0/3; 12.0002	3/3; 4.3099
003	CommonStart	3/3; 4.7903	3/3; 4.6932
042	Cold	3/3; 0.3284	3/3; 0.4928
042	CommonStart	3/3; 0.0000	3/3; 0.0000

The common start removes the completion advantage of GM in both favorable cases. In 001, paired GM/Base work ratios are 1.0555, 1.0519 and 1.3686; GM requires more work in all three pairs. In 003 the ratios are 0.9957, 1.1114 and 0.7683, with mixed signs. The zero native-work reports for warm 042 do not mean zero total cost and are not divided into ratios. No root LP is logged there.

Root objectives match at printed precision within all 15 Base/GM pairs where root relaxations are logged. The start changes some root objectives relative to Cold: it can alter objective cutoffs, presolve and later search. Therefore the intervention supports dependence on incumbent access, without isolating one heuristic or proving that all effects have that cause.

3. Acquisition and exact checks

The references are offline diagnostic inputs. Their patterns came from previous GM runs; the new exact LP construction/check took 1.79–1.84 seconds per case, separate from the earlier work needed to obtain those patterns. This is not a deployable algorithm with free starts, nor a total-time speedup.

All 15 unique final integer patterns entered a queue fixed before posthoc LP checking (five-second limit, no replacement). Thirteen have exact primal/dual optima. Two remain unknown, both from Base-Cold 001, which did not reach the requested gap. Warm comparisons have certified continuations at comparable costs. Conditional pattern optima are not global MILP lower bounds.

No reported bound exceeds the common exact feasible objective in this experiment. Absence of contradiction is not validation of global bounds. All native candidates and logs are retained. Original-row checks use stored binary64 coefficients and rational arithmetic after integer rounding; they do not certify an intended pre-rounding physical model.

The independent replay checks 1,179,324 exact products, 226 candidate objectives, all 36 final points and pattern assignments, all 18 loaded common starts, three reference LPs and 15 final-pattern LPs. Four tests passed before MILP calls.

4. What this contributes to OMS

The result directly tests a mechanism behind the paper's observed operational effects. It supports a short qualification of existing evidence, rather than another principal contribution or a claim that scaling strengthens the LP. No extra campaign is warranted by these three known cases.

Concrete candidate paragraph:

An auxiliary MPS-based diagnostic with power-of-two GM factors further limits mechanism attribution. On three known dispatch cases with three new seeds, Base and GM receive the same independently checked feasible reference as a MIP start. In the two cases where cold-start GM reaches the requested native gap in 3/3 calls while Base reaches it in 0/3, the common start makes both reach it in 3/3. GM then requires more work in all three pairs on one case and has mixed work differences on the other. This diagnostic conditions on an offline reference; it is not a new solution method or a global optimality certificate. It shows that the observed completion advantage depends on incumbent access and cannot be attributed to exported-matrix conditioning alone; it does not isolate a particular heuristic.

5. Limits and reproducibility

One application family, three selected cases, one solver/version and short limits cannot establish a general ranking. A common start affects the whole solver pipeline. Its acceptance is checked because the Gurobi Start API does not guarantee that a supplied vector becomes an incumbent. The incumbent source is fully declared and is not used to train a predictive rule.

The package contains canonical inputs, source provenance, archived discovery records, exact references, schedules, native logs, candidates, fixed-pattern LPs, rational certificates and solver-free replay. Python, NumPy and pytest suffice for checks; pdfLaTeX builds this note. New solves additionally require the declared Gurobi and GMP-enabled SoPlex binaries/licences. The manuscript, frozen OMS v1.3.5 delivery and eleven previous research archives are preserved.